

## 3.7 Compliant, data-driven city of Trondheim (MUNDAT)

### 3.7.1 Case Description

The MUNDAT use case focuses on efficient and compliant sharing of municipal data to data consumers (the public, academia, industry, other governmental institutes and internally). The use case encompasses activities from a data sharing request is issued from a data consumer and until relevant datasets are shared by the municipality. During this process several compliance activities and data processing operations are required, some of which may in the future be performed using automated or semi-automated tool support provided by DataPACT.

Today, Trondheim municipality receives many requests to supply datasets, e.g., for academic or industrial research or data innovation purposes. Requests are currently directed to the municipality's numerous contact points. There is no “single window” for data requests, making it difficult to assess their legal implications and boundaries in a consistent manner. This combined with a lack of technical data management infrastructure and associated routines makes it difficult to efficiently assemble the requested data correctly and according to a comprehensive compliance framework. Sharing municipal data, either as anonymised (aggregated) datasets through our open data portal, or as protected datasets subject to formal data sharing contracts, requires complex pipeline steps with proper legal, ethical and social impact assessment. Furthermore, the compliance assessment process typically requires much effort and different personnel needs, making it difficult to upscale the data delivery process. Figure 7 shows a schematic and high-level view of the activities involved in each data sharing process.

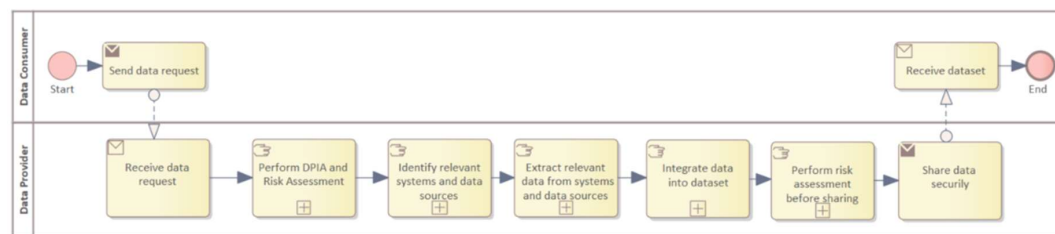


Figure 7: Workflow of activities in the MUNDAT case.

The main activities are further described in Table 37 below.

Table 37: Key activities of the MUNDAT use case.

| Activity                                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Receive data request                       | Data sharing requests are received by different entities in the municipality and from various sources (research, student activities, media, private companies and other public sector organisations).                                                                                                                                                                                                                                              |
| Perform DPIA and risk assessment           | Upon reception of a data sharing request, a DPIA process is initiated to ensure that the data processing is compliant with law and defined procedures. Risk assessment must be performed at different stages during the process flow. Risk assessments are performed early on (more high-level assessments), before extracting data from data sources, after integrating data from different sources into a final dataset and before sharing data. |
| Identify relevant systems and data sources | Relevant data must be identified in the over 300 different systems maintained by the municipality.                                                                                                                                                                                                                                                                                                                                                 |

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|                                                     |                                                                                                                                                    |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Extract relevant data from systems and data sources | Once identified, relevant data must be extracted from identified systems. Some of these systems have APIs, others have not.                        |
| Integrate data into dataset                         | Data from disparate sources are integrated into a dataset. The activity typically involves data interpretation, data cleaning and harmonisation.   |
| Perform risk assessment before sharing              | Often, data within the dataset are subject to anonymisation (as long as the dataset is to be provided as open data).                               |
| Share data securely                                 | If the data are not open and can be put on the municipality's open data platform, the data must be shared to the data consumer in a secure manner. |

In order to ensure that these activities are conducted in an efficient and compliant manner, the main objectives of the MUNDAT case in DataPACT include:

- Implement a single contact point and associated processes for managing data sharing requests in the municipality.
- Establish data pipelines within our data management infrastructure that will make the process of matching data sharing requests with available data more efficient.
- Enable more efficient compliance processes by automating activities when possible and implementing a coherent compliance framework.

To address these objectives, the municipality of Trondheim will have to extend existing processes and our current data management platform with frameworks and tool support developed in DataPACT.

### 3.7.2 Users Description

The MUNDAT use case encompasses several stakeholder categories, each playing a key role in various stages of the pipeline, from data collection via compliance assurance to data delivery. These are the main roles within the scope of this business case in DataPACT:

- **Data Provider:** Makes datasets available, either as open data or under controlled access, ensuring secure and compliant sharing.
- **Data Consumer:** Requests and uses the data.
- **Data Protection Officer:** Must be consulted during a DPIA process but has no formal authority to decide whether data can be shared or not.
- **Compliance Process Owner:** Responsible for ensuring that data sharing (including data processing) is performed in compliance with laws.
- **Data Engineer:** Responsible for the data processing and technical arrangements related to the data processing and data sharing process.
- **Information Security Expert:** Responsible for ensuring that the technical infrastructure as well as the data processing and sharing processes adhere to security standards and practices.
- **Service Owner:** Formally responsible for a system holding data.

Not all stakeholder roles are represented with dedicated user stories, as some mainly provide oversight, expertise, or governance rather than directly interacting with the system.

### 3.7.3 Compliance Challenges

The MUNDAT use case introduces specific compliance challenges. With the business case focus on channelling data sharing requests through a single window and optimising pipeline processes to ensure compliant data processing and provision these are some of the challenges we currently experience:

- **No systematic approach for managing data requests:** With the missing overview of how and by whom different data requests are managed, we cannot guarantee that data processing and data delivery is processed in a compliant manner. What is needed is a “single window” access point and routines that channel the requests to the appropriate recipients within the municipality so that the requests and associated data processing operations are managed in an efficient and compliant manner.
- **Efficiency in compliant data processing and provision:** It's currently challenging to assess compliance issues and risks at a detailed data level throughout the data lifecycle, from extraction and integration to sharing. This makes it difficult to efficiently comply with GDPR and forthcoming Norwegian laws based on the Data Governance Act, the Open Data Directive, the Data Act and the AI Act. The challenges stem from several factors: a limited set of employees know how to fully execute Data Protection Impact Assessments (DPIAs), uncertainty about which data sources are subject to compliance processes, a lack of data dictionaries or metadata to define compliance requirements, and the absence of "compliance metadata" annotations. Furthermore, integrating data from various sources complicates the ability to identify re-identifiable Personally Identifiable Information (PII) and other sensitive data. Finally, data sharing currently requires top-level management approval, necessitating improved documentation of compliance measures taken during data processing and high-level descriptions of prepared datasets.

### 3.7.4 Datasets

Table 38 compares the core characteristics of the two datasets used in MUNDAT.

*Table 38: Datasets used in the MUNDAT case.*

| Dataset Name     | Description                                                                                                                                                                                                                                                                                                                                                           | Data Type               | Data Size and Format                      | Personal Data (Y/N) | Licensing               |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-------------------------------------------|---------------------|-------------------------|
| <b>Sickleave</b> | This dataset is composed of data from different systems, including a system for registering sick leave in job rotation units, data from work environment surveys, turnover reports and demographic data (e.g., employee age distribution) from the HR system. This dataset is targeted for research aiming to identify contributing factors and possible measures for | Periodic data (monthly) | ~1 MB (aggregated and anonymised version) | Yes, anonymised     | Public research dataset |

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|            |                                                                                                                                                                                                                                      |             |        |    |              |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------|----|--------------|
|            | reducing the high sick leave in the municipality.                                                                                                                                                                                    |             |        |    |              |
| <b>IoT</b> | This dataset contains sensor data from sensors measuring the level of sand in stormwater basins. The dataset consists of dynamic real-time data that when combined with other municipal data can inform data-driven decision making. | Time series | ~50 MB | No | Open dataset |

## 3.7.5 User Stories

Table 39: User stories in the MUNDAT case.

| ID       | Use Case Name                  | As a <type of user>      | I want to <goal/objective>                                                            | So that <benefit/result/some reason>                                                        | Relevant DataPACT Result |
|----------|--------------------------------|--------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|--------------------------|
| TK.US.01 | Process dataset request        | Compliance Process Owner | assess whether data request for open or protected data is in line with compliant data | I can ensure that datasets are shared legally and ethically                                 | AssessR R5.4             |
| TK.US.02 | Assess compliance requirements | Compliance Process Owner | have automated support in identifying legal references and ethical guidelines         | I can justify data sharing decisions                                                        | AssessR R5.4             |
| TK.US.03 | Ensure secure dataset delivery | Compliance Process Owner | specify access control and use policies                                               | I can ensure that datasets I share are treated according to contracts and routines          | PolicyR R2.1             |
| TK.US.04 | Ensure secure dataset delivery | Compliance Process Owner | manage data sharing contracts                                                         | responsibilities related to legal compliance, ethical considerations and trust are declared | PolicyR R2.3             |
| TK.US.05 | Identify relevant data sources | Compliance Process Owner | be able to efficiently collect consent from relevant parties (if relevant/possible)   | I ensure that informed consent for processing and sharing the data is in place              | PolicyR R2.2             |

Deliverable D5.1: Use cases requirements analysis and specifications

|          |                                                     |                          |                                                                                                                                          |                                                                                   |                                                                 |
|----------|-----------------------------------------------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------|
| TK.US.06 | Extract relevant data / integrate data into dataset | Data Engineer            | design the complete data / AI pipeline                                                                                                   | I can efficiently manage data flow and processing of data sharing requests        | PipelineR<br>R1.2, R1.3, R1.4<br><br>GreenR<br>R4.1, R4.2, R4.3 |
| TK.US.07 | Extract relevant data / integrate data into dataset | Data Engineer            | use visual interfaces for easy configuration and monitoring of operations                                                                | I can easily track data lineage and traceability                                  | PipelineR<br>R1.2, R1.3, R1.4                                   |
| TK.US.08 | Ensure secure dataset delivery                      | Data Provider            | securely make data available through an established dataspace                                                                            | authorized users can find and access the data                                     | PolicyR<br>R2.1<br><br>ApplyR<br>R6.3<br><br>PipelineR<br>R1.1  |
| TK.US.09 | Receive dataset request                             | Data Consumer            | adhere to relevant legislation without having to read law books                                                                          | get access to data in a legally compliant manner                                  | AssessR<br>R5.4                                                 |
| TK.US.10 | Publish approved datasets in data catalog           | Data Consumer            | be able to access a data repository with pre-authorized and compliant datasets                                                           | I can access and use data and provide my services                                 | Internal development                                            |
| TK.US.11 | Determine if a full DPIA is needed                  | Compliance Process Owner | automated support in making a decision as to whether a full DPIA is needed                                                               | the initial part of the compliance management process can be made more efficient. | AssessR<br>R5.4, R5.5                                           |
| TK.US.12 | Perform risk assessment                             | Data Engineer            | have automated support in detecting risks for direct or indirect identification of PII or other sensitive data in raw and processed data | data are compliant with relevant laws and regulations                             | PipelineR<br>R1.1<br><br>ApplyR<br>R6.3<br><br>AssessR<br>R5.4  |
| TK.US.13 | Perform data protection                             | Data Engineer            | have tool support for protecting (aggregating/anonymising, masking,                                                                      | direct or indirect identification of sensitive data cannot be made                | Internal development                                            |

#### Deliverable D5.1: Use cases requirements analysis and specifications

|          |                                      |                  |                                                                                                                                         |                                                                                                                     |                         |
|----------|--------------------------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------|
|          |                                      |                  | pseudonymising)<br>data that needs to<br>be protected                                                                                   |                                                                                                                     |                         |
| TK.US.14 | Ensure secure<br>dataset<br>delivery | Data<br>Engineer | have automated<br>support in<br>documenting the<br>data and how it<br>has been<br>processed during<br>the construction of<br>a dataset. | those making a<br>decision on<br>whether the<br>dataset can be<br>shared can trust<br>that compliance<br>is assured | PipelineR<br>R1.3, R1.4 |

### 3.7.6 Functional Requirements

Table 40: Functional requirements in the MUNDAT case.

| Requirement ID | Description                                                                                                                                                                                                      | Source             | Priority    |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------|
| TK.FR.01       | Available datasets shall be described using a standardised metadata vocabulary                                                                                                                                   | TK.US.09, TK.US.10 | Must have   |
| TK.FR.02       | A secure data catalog / dataspace shall be provided for sharing datasets                                                                                                                                         | TK.US.08           | Could have  |
| TK.FR.03       | A tool that automatically identifies compliance issues on the basis of a detailed data need description shall be developed                                                                                       | TK.US.01           | Must have   |
| TK.FR.04       | A tool that automatically identifies potential compliance issues on the basis of an established dataset / collection of data shall be developed.                                                                 | TK.US.02, TK.US.11 | Must have   |
| TK.FR.05       | A metadata vocabulary for describing data lineage / traceability of data and data processing operations shall be developed                                                                                       | TK.US.07           | Should have |
| TK.FR.06       | It shall be possible to track data lineage / traceability of data and data processing operations using a visual interface                                                                                        | TK.US.07           | Should have |
| TK.FR.07       | A tool for managing (creating, storing, monitoring) consent related to PII data shall be developed                                                                                                               | TK.US.05           | Could have  |
| TK.FR.08       | GUI for designing a complete data pipeline on the basis of metadata describing data source, data profile (including PII and other sensitivity characteristics), access and usage control, lineage, formats, etc. | TK.US.06           | Should have |
| TK.FR.09       | Develop tooling for automatically protecting (anonymising,                                                                                                                                                       | TK.US.13           | Could have  |

#### Deliverable D5.1: Use cases requirements analysis and specifications

|          |                                                                                                                                              |                    |             |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------|
|          | pseudonymising) sensitive data                                                                                                               |                    |             |
| TK.FR.10 | Develop tool support for automatic identification and tagging of PII-data from a data dump                                                   | TK.US.12           | Should have |
| TK.FR.11 | A tool for user-friendly specification and generation of access control and usage policies of a dataset shall be developed.                  | TK.US.03           | Could have  |
| TK.FR.12 | Develop a tool that automatically generates a report describing data and how it has been processed during dataset construction               | TK.US.14           | Should have |
| TK.FR.13 | Configuration of access and usage control to datasets shared in a secure dataspace shall be made possible through a user-friendly interface. | TK.US.03, TK.US.08 | Could have  |
| TK.FR.14 | A tool for managing (creating, storing, monitoring) data sharing contracts shall be developed                                                | TK.US.04           | Could have  |
| TK.FR.15 | A user-friendly interface (connector) shall provide functionality for sharing a dataset in a secure dataspace                                | TK.US.08           | Could have  |

### 3.7.7 Non-Functional Requirements

Table 41: Non-functional requirements in the MUNDAT case.

| Requirement ID | Description                                                                                                  | Source   | Priority   |
|----------------|--------------------------------------------------------------------------------------------------------------|----------|------------|
| TK.NFR.01      | Secure security protocols / encryption shall ensure that datasets can be secured both at rest and in transit | TK.US.08 | Could have |

### 3.7.8 KPIs

Table 42: Key Performance Indicators in the MUNDAT case.

| ID | Key Performance Indicator                        | Measure                                                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----|--------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Faster data request-to data provision lead time. | 50% reduced fulfilment time for providing data upon request | <p>Early assessment of whether a data sharing process requires a full DPIA can be reduced by implementing the LLM-assisting tool for understanding compliance requirements from a standardised description of the data request (AssessR R5.4).</p> <p>Reducing the time spent on identifying relevant data sources mapping to the data request. This process initially requires annotating existing data with metadata (internal development), but introducing methodology and tools for managing</p> |

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**Deliverable D5.1: Use cases requirements analysis and specifications**

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|---|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |                                                                                                               |                                                                              | <p>compliant data pipelines as provided by DataPACT (PipelineR, AssessR) will support this process.</p> <p>Explainability and traceability of existing data and datasets will increase data quality and trust in data, contributing to that much effort that is currently spent on manual interpretation of data provenance and processing will be reduced. Relevant functionality is provided by DataPACT (PipelineR R1.4).</p> |
| 2 | One data sharing request contact point to streamline data processing, compliance management and data sharing. | Channel the currently vast number of contact points to 1 single entry point. | Implement with built-in tools to channel the requests complying with legislation, with significant time reduction for manual resource use.                                                                                                                                                                                                                                                                                       |